

PANTHER JUNCTION 2 N, TX, USA

WMO: 747280

Lat: 29.3483N

Lon: 103.2093W

Elev: 3494

StdP: 12.93

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 22016

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 26.1	(c) 29.5	(d) -1.8	(e) 5.7	(f) 62.0	(g) 2.5	(h) 7.1	(i) 62.7	(j) 12.9	(k) 66.9	(l) 11.1	(m) 63.3	(n) 3.2	(o) N/A	(p) 0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 25.7	(c) 100.3	(d) 63.9	(e) 98.3	(f) 63.9	(g) 96.2	(h) 63.8	(i) 70.4	(j) 84.2	(k) 69.4	(l) 83.5	(m) 68.8	(n) 83.5	(o) 5.4	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	116.2	73.1	66.3	110.2	73.2	65.0	105.2	73.7	36.6	83.6	35.8	83.0	35.1	83.1	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.0	(b) 9.8	(c) 8.7	(e) 19.8	(f) 104.7	(g) 4.9	(h) 1.8	(i) 16.3	(j) 105.9	(k) 13.4	(l) 107.0	(m) 10.6	(n) 108.0	(o) 7.0	(p) 109.3		
(a) 11.0	(b) 9.8	(c) 8.7	(e) 17.2	(f) 72.1	(g) 5.0	(h) 0.9	(i) 13.5	(j) 72.8	(k) 10.6	(l) 73.3	(m) 7.8	(n) 73.8	(o) 4.1	(p) 74.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.8	49.4	55.5	62.2	69.8	76.6	82.7	82.0	82.1	75.8	68.7	57.5	51.1
	DBStd	13.80	8.68	9.72	8.57	7.01	6.87	4.52	4.08	4.33	5.02	7.27	8.23	8.08
	HDD50	301	118	51	14	1	0	0	0	0	0	2	29	86
	HDD65	1687	486	281	151	34	8	0	0	0	3	49	244	431
	CDD50	6813	99	205	393	596	826	980	992	994	773	580	253	122
	CDD65	2724	2	16	64	178	369	530	527	529	326	163	18	2
	CDH74	34728	64	356	1125	2563	4796	7087	6435	6717	3291	1874	349	71
	CDH80	17295	4	85	355	1128	2539	4047	3381	3588	1397	706	63	2
(14) Wind	WSAvg	4.1	3.6	3.9	4.2	4.7	4.9	4.8	4.7	4.1	3.7	3.6	3.5	3.5
(15) Precipitation	PrecAvg	13.2	0.5	0.5	0.4	0.5	1.3	1.7	2.6	2.3	1.5	1.2	0.6	0.4
	PrecMax	20.6	2.7	2.3	2.1	1.7	3.5	4.1	5.0	6.3	3.7	4.6	2.2	2.6
	PrecMin	7.0	0.0	0.0	0.0	0.0	0.0	0.3	0.6	0.5	0.0	0.0	0.0	0.0
	PrecStd	3.4	0.6	0.6	0.5	0.5	0.7	1.0	1.4	1.4	1.2	1.3	0.6	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	87.5	90.6	96.0	101.8	104.3	100.7	100.7	96.1	92.2	85.5	79.8
		MCWB	51.9	53.9	54.9	56.8	61.3	63.0	65.6	66.2	64.4	60.8	56.1	52.7
	2%	DB	75.4	82.5	87.2	92.9	98.1	101.5	98.2	98.1	93.1	89.9	81.6	76.0
		MCWB	50.1	52.7	54.3	56.1	59.3	63.6	65.3	65.7	63.9	61.3	55.2	51.1
	5%	DB	71.5	78.3	84.3	90.1	95.3	98.9	96.1	96.2	90.7	87.3	78.1	72.5
		MCWB	48.9	51.4	53.6	55.2	58.7	63.9	65.1	65.6	64.2	61.0	54.8	50.9
	10%	DB	67.2	74.6	80.9	87.0	92.6	96.2	93.8	94.2	88.1	84.3	74.7	68.5
		MCWB	47.3	50.6	53.1	54.9	59.4	64.5	65.3	65.6	64.2	60.5	54.3	49.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	58.6	62.1	64.8	69.6	71.1	72.3	71.3	70.6	68.6	62.9	56.3
		MCDB	65.1	75.5	77.6	80.5	86.6	86.4	84.1	87.4	82.3	83.3	72.7	70.1
	2%	WB	53.0	56.5	59.5	62.6	68.0	69.5	70.3	69.9	69.4	66.9	60.4	54.2
		MCDB	69.6	73.2	74.7	77.7	84.2	85.7	82.0	85.2	82.2	80.3	71.9	69.2
	5%	WB	50.5	54.1	57.5	61.2	66.4	68.6	69.2	69.0	68.7	65.4	58.4	52.5
		MCDB	69.3	71.6	74.0	76.8	82.1	85.6	81.9	84.5	81.2	78.5	71.3	67.3
	10%	WB	48.2	52.0	55.8	59.4	65.2	67.8	68.3	68.2	68.0	63.9	56.0	50.3
		MCDB	65.8	71.7	72.8	75.5	80.7	85.5	82.9	84.6	80.3	76.8	70.4	64.8
(35) Mean Daily Temperature Range	5% DB	MDBR	26.3	28.7	29.5	29.2	27.1	25.7	22.1	22.2	21.5	26.0	26.4	25.9
		MCDBR	31.8	33.2	33.9	32.1	29.7	28.5	24.9	24.5	25.6	28.8	31.0	31.1
		MCWBR	14.9	14.2	13.5	12.4	12.1	7.8	6.3	6.2	7.2	10.4	13.4	13.9
		MCDBR	29.9	31.0	29.7	28.3	27.1	24.9	20.6	22.0	19.4	23.2	27.0	27.7
	5% WB	MCWBR	14.8	14.2	11.8	11.2	11.4	7.1	6.0	6.1	5.7	8.9	13.0	13.4
(40) Clear-Sky Solar Irradiance	taub		0.262	0.267	0.281	0.319	0.352	0.367	0.390	0.366	0.358	0.302	0.277	0.263
	taud		2.659	2.627	2.570	2.441	2.383	2.417	2.393	2.482	2.437	2.611	2.616	2.634
	Ebn at Noon		317	322	321	309	297	290	284	291	291	305	307	311
	Edh at Noon		24	27	31	36	39	37	38	35	35	27	25	24
(44) All-Sky Solar Radiation	RadAvg		1204	1519	1904	2210	2352	2358	2178	2063	1797	1569	1289	1123
	RadStd		113	132	95	122	98	105	128	128	115	131	110	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+401	+277

Nomenclature: See separate page